

Training 初始值参数余量的测试方式

1, dump 回标

进入 uboot, 输入 ddr_test_cmd 0x37,从 log 中找到 read_dqs_gate_delay/read_dqs_delay/read_dq_bit_delay 参数回填到 timing.c 文件中.

Timing.c

```
.cfg_ddr_training_delay_ps.read_dqs_gate_delay[0]=0x000001a4, // 420
.cfg_ddr_training_delay_ps.read_dqs_gate_delay[1]=0x0000001c, // 455
.cfg_ddr_training_delay_ps.read_dqs_gate_delay[2]=0x000001d9, // 473
.cfg_ddr_training_delay_ps.read_dqs_gate_delay[3]=0x000001f4, // 500
.cfg_ddr_training_delay_ps.read_dqs_gate_delay[4]=0x000001d8, // 472
.cfg_ddr_training_delay_ps.read_dqs_gate_delay[5]=0x000001fb, // 507
.cfg_ddr_training_delay_ps.read_dqs_gate_delay[6]=0x000001d6, // 472
.cfg_ddr_training_delay_ps.read_dqs_gate_delay[7]=0x000001f4, // 500
.cfg_ddr_training_delay_ps.read_dqs_delay[0]=0x00000079, // 121
.cfg_ddr_training_delay_ps.read_dqs_delay[1]=0x0000007a, // 122
.cfg_ddr_training_delay_ps.read_dqs_delay[2]=0x00000085, // 133
.cfg_ddr_training_delay_ps.read_dqs_delay[3]=0x00000065, // 101
.cfg_ddr_training_delay_ps.read_dqs_delay[4]=0x00000080, // 128
.cfg_ddr_training_delay_ps.read_dqs_delay[5]=0x00000077, // 119
.cfg_ddr_training_delay_ps.read_dqs_delay[6]=0x0000008e, // 140
.cfg_ddr_training_delay_ps.read_dq_bit_delay[0]=0x00000073, // 115
.cfg_ddr_training_delay_ps.read_dq_bit_delay[1]=0x0000003f, // 63
.cfg_ddr_training_delay_ps.read_dq_bit_delay[2]=0x00000058, // 88
.cfg_ddr_training_delay_ps.read_dq_bit_delay[3]=0x00000057, // 87
.cfg_ddr_training_delay_ps.read_dq_bit_delay[4]=0x00000042, // 66
.cfg_ddr_training_delay_ps.read_dq_bit_delay[5]=0x00000031, // 51
.cfg_ddr_training_delay_ps.read_dq_bit_delay[6]=0x00000040, // 77
.cfg_ddr_training_delay_ps.read_dq_bit_delay[7]=0x00000042, // 66
.cfg_ddr_training_delay_ps.read_dq_bit_delay[8]=0x00000032, // 50
.cfg_ddr_training_delay_ps.read_dq_bit_delay[9]=0x00000044, // 68
.cfg_ddr_training_delay_ps.read_dq_bit_delay[10]=0x00000036, // 62
.cfg_ddr_training_delay_ps.read_dq_bit_delay[11]=0x00000031, // 63
.cfg_ddr_training_delay_ps.read_dq_bit_delay[12]=0x00000046, // 70
.cfg_ddr_training_delay_ps.read_dq_bit_delay[13]=0x00000033, // 51
.cfg_ddr_training_delay_ps.read_dq_bit_delay[14]=0x00000049, // 73
.cfg_ddr_training_delay_ps.read_dq_bit_delay[15]=0x00000036, // 62
.cfg_ddr_training_delay_ps.read_dq_bit_delay[16]=0x00000055, // 85
.cfg_ddr_training_delay_ps.read_dq_bit_delay[17]=0x00000043, // 67
.cfg_ddr_training_delay_ps.read_dq_bit_delay[18]=0x00000051, // 87
.cfg_ddr_training_delay_ps.read_dq_bit_delay[19]=0x00000038, // 56
.cfg_ddr_training_delay_ps.read_dq_bit_delay[20]=0x00000041, // 65
.cfg_ddr_training_delay_ps.read_dq_bit_delay[21]=0x00000030, // 61
.cfg_ddr_training_delay_ps.read_dq_bit_delay[22]=0x00000039, // 56
.cfg_ddr_training_delay_ps.read_dq_bit_delay[23]=0x0000003b, // 59
.cfg_ddr_training_delay_ps.read_dq_bit_delay[24]=0x00000046, // 70
.cfg_ddr_training_delay_ps.read_dq_bit_delay[25]=0x00000035, // 53
.cfg_ddr_training_delay_ps.read_dq_bit_delay[26]=0x0000003f, // 63
.cfg_ddr_training_delay_ps.read_dq_bit_delay[27]=0x00000037, // 55
.cfg_ddr_training_delay_ps.read_dq_bit_delay[28]=0x00000034, // 52
.cfg_ddr_training_delay_ps.read_dq_bit_delay[29]=0x00000041, // 65
.cfg_ddr_training_delay_ps.read_dq_bit_delay[30]=0x0000003a, // 58
.cfg_ddr_training_delay_ps.read_dq_bit_delay[31]=0x0000003b, // 59
.cfg_ddr_training_delay_ps.read_dq_bit_delay[32]=0x00000034, // 52
.cfg_ddr_training_delay_ps.read_dq_bit_delay[33]=0x00000057, // 87
.cfg_ddr_training_delay_ps.read_dq_bit_delay[34]=0x00000059, // 89
.cfg_ddr_training_delay_ps.read_dq_bit_delay[35]=0x00000037, // 63
.cfg_ddr_training_delay_ps.read_dq_bit_delay[36]=0x00000044, // 70
.cfg_ddr_training_delay_ps.read_dq_bit_delay[37]=0x00000066, // 102
.cfg_ddr_training_delay_ps.read_dq_bit_delay[38]=0x00000067, // 103
.cfg_ddr_training_delay_ps.read_dq_bit_delay[39]=0x0000004e, // 78
.cfg_ddr_training_delay_ps.read_dq_bit_delay[40]=0x0000003b, // 59
.cfg_ddr_training_delay_ps.read_dq_bit_delay[41]=0x00000058, // 88
.cfg_ddr_training_delay_ps.read_dq_bit_delay[42]=0x00000043, // 77
.cfg_ddr_training_delay_ps.read_dq_bit_delay[43]=0x0000003d, // 61
.cfg_ddr_training_delay_ps.read_dq_bit_delay[44]=0x00000050, // 80
.cfg_ddr_training_delay_ps.read_dq_bit_delay[45]=0x00000044, // 68
.cfg_ddr_training_delay_ps.read_dq_bit_delay[46]=0x00000042, // 66
.cfg_ddr_training_delay_ps.read_dq_bit_delay[47]=0x0000004b, // 75
.cfg_ddr_training_delay_ps.read_dq_bit_delay[48]=0x00000036, // 54
.cfg_ddr_training_delay_ps.read_dq_bit_delay[49]=0x00000041, // 79
.cfg_ddr_training_delay_ps.read_dq_bit_delay[50]=0x00000047, // 67
.cfg_ddr_training_delay_ps.read_dq_bit_delay[51]=0x00000044, // 77
.cfg_ddr_training_delay_ps.read_dq_bit_delay[52]=0x00000058, // 88
.cfg_ddr_training_delay_ps.read_dq_bit_delay[53]=0x00000047, // 71
.cfg_ddr_training_delay_ps.read_dq_bit_delay[54]=0x0000004b, // 79
.cfg_ddr_training_delay_ps.read_dq_bit_delay[55]=0x0000003b, // 59
.cfg_ddr_training_delay_ps.read_dq_bit_delay[56]=0x00000045, // 69
.cfg_ddr_training_delay_ps.read_dq_bit_delay[57]=0x00000043, // 67
.cfg_ddr_training_delay_ps.read_dq_bit_delay[58]=0x0000003e, // 62
.cfg_ddr_training_delay_ps.read_dq_bit_delay[59]=0x00000042, // 66
.cfg_ddr_training_delay_ps.read_dq_bit_delay[60]=0x00000040, // 72
.cfg_ddr_training_delay_ps.read_dq_bit_delay[61]=0x00000037, // 55
.cfg_ddr_training_delay_ps.read_dq_bit_delay[62]=0x00000043, // 67
.cfg_ddr_training_delay_ps.read_dq_bit_delay[63]=0x00000034, // 63
.cfg_ddr_training_delay_ps.read_dq_bit_delay[64]=0x0000003c, // 60
.cfg_ddr_training_delay_ps.read_dq_bit_delay[65]=0x00000049, // 73
.cfg_ddr_training_delay_ps.read_dq_bit_delay[66]=0x00000043, // 67
.cfg_ddr_training_delay_ps.read_dq_bit_delay[67]=0x00000036, // 54
.cfg_ddr_training_delay_ps.read_dq_bit_delay[68]=0x00000037, // 55
.cfg_ddr_training_delay_ps.read_dq_bit_delay[69]=0x00000050, // 93
.cfg_ddr_training_delay_ps.read_dq_bit_delay[70]=0x00000060, // 96
.cfg_ddr_training_delay_ps.read_dq_bit_delay[71]=0x00000046, // 70
```

log: log 中[0]需要去掉才能回填到 timing.c 中

```
[20240405 20:58:25].cfg_ddr_training_delay_ps.read_dqs_gate_delay[0]=0x0000007f, // 127
[20240405 20:58:25].cfg_ddr_training_delay_ps.read_dqs_gate_delay[1]=0x00000091, // 145
[20240405 20:58:25].cfg_ddr_training_delay_ps.read_dqs_gate_delay[2]=0x00000081, // 129
[20240405 20:58:25].cfg_ddr_training_delay_ps.read_dqs_gate_delay[3]=0x00000083, // 131
[20240405 20:58:25].cfg_ddr_training_delay_ps.read_dqs_gate_delay[4]=0x000000a2, // 162
[20240405 20:58:25].cfg_ddr_training_delay_ps.read_dqs_gate_delay[5]=0x000000a0, // 160
[20240405 20:58:25].cfg_ddr_training_delay_ps.read_dqs_gate_delay[6]=0x0000009c, // 156
[20240405 20:58:25].cfg_ddr_training_delay_ps.read_dqs_gate_delay[7]=0x000000ae, // 174
[20240405 20:58:25].cfg_ddr_training_delay_ps.read_dqs_delay[0]=0x0000007b, // 123
[20240405 20:58:25].cfg_ddr_training_delay_ps.read_dqs_delay[1]=0x0000006d, // 109
[20240405 20:58:25].cfg_ddr_training_delay_ps.read_dqs_delay[2]=0x00000074, // 116
[20240405 20:58:25].cfg_ddr_training_delay_ps.read_dqs_delay[3]=0x00000073, // 115
[20240405 20:58:25].cfg_ddr_training_delay_ps.read_dqs_delay[4]=0x00000081, // 129
[20240405 20:58:25].cfg_ddr_training_delay_ps.read_dqs_delay[5]=0x00000076, // 118
[20240405 20:58:25].cfg_ddr_training_delay_ps.read_dqs_delay[6]=0x0000007a, // 122
[20240405 20:58:25].cfg_ddr_training_delay_ps.read_dqs_delay[7]=0x00000073, // 115
[20240405 20:58:25].cfg_ddr_training_delay_ps.read_dq_bit_delay[0]=0x00000054, // 84
[20240405 20:58:25].cfg_ddr_training_delay_ps.read_dq_bit_delay[1]=0x00000053, // 83
[20240405 20:58:25].cfg_ddr_training_delay_ps.read_dq_bit_delay[2]=0x00000042, // 66
[20240405 20:58:25].cfg_ddr_training_delay_ps.read_dq_bit_delay[3]=0x0000003b, // 59
[20240405 20:58:25].cfg_ddr_training_delay_ps.read_dq_bit_delay[4]=0x0000004e, // 79
[20240405 20:58:25].cfg_ddr_training_delay_ps.read_dq_bit_delay[5]=0x00000034, // 52
[20240405 20:58:25].cfg_ddr_training_delay_ps.read_dq_bit_delay[6]=0x00000043, // 67
[20240405 20:58:25].cfg_ddr_training_delay_ps.read_dq_bit_delay[7]=0x0000003d, // 61
[20240405 20:58:25].cfg_ddr_training_delay_ps.read_dq_bit_delay[8]=0x00000044, // 68
[20240405 20:58:25].cfg_ddr_training_delay_ps.read_dq_bit_delay[9]=0x00000045, // 69
[20240405 20:58:25].cfg_ddr_training_delay_ps.read_dq_bit_delay[10]=0x00000042, // 66
[20240405 20:58:25].cfg_ddr_training_delay_ps.read_dq_bit_delay[11]=0x00000032, // 50
[20240405 20:58:25].cfg_ddr_training_delay_ps.read_dq_bit_delay[12]=0x00000046, // 70
[20240405 20:58:25].cfg_ddr_training_delay_ps.read_dq_bit_delay[13]=0x00000049, // 73
[20240405 20:58:25].cfg_ddr_training_delay_ps.read_dq_bit_delay[14]=0x00000034, // 61
[20240405 20:58:25].cfg_ddr_training_delay_ps.read_dq_bit_delay[15]=0x00000035, // 53
[20240405 20:58:25].cfg_ddr_training_delay_ps.read_dq_bit_delay[16]=0x00000048, // 68
[20240405 20:58:25].cfg_ddr_training_delay_ps.read_dq_bit_delay[17]=0x00000043, // 67
[20240405 20:58:25].cfg_ddr_training_delay_ps.read_dq_bit_delay[18]=0x00000045, // 69
[20240405 20:58:25].cfg_ddr_training_delay_ps.read_dq_bit_delay[19]=0x0000003c, // 60
[20240405 20:58:25].cfg_ddr_training_delay_ps.read_dq_bit_delay[20]=0x00000059, // 89
[20240405 20:58:25].cfg_ddr_training_delay_ps.read_dq_bit_delay[21]=0x00000053, // 83
[20240405 20:58:25].cfg_ddr_training_delay_ps.read_dq_bit_delay[22]=0x00000032, // 50
[20240405 20:58:25].cfg_ddr_training_delay_ps.read_dq_bit_delay[23]=0x00000037, // 55
[20240405 20:58:25].cfg_ddr_training_delay_ps.read_dq_bit_delay[24]=0x00000048, // 72
[20240405 20:58:25].cfg_ddr_training_delay_ps.read_dq_bit_delay[25]=0x00000044, // 77
[20240405 20:58:25].cfg_ddr_training_delay_ps.read_dq_bit_delay[26]=0x00000043, // 67
[20240405 20:58:25].cfg_ddr_training_delay_ps.read_dq_bit_delay[27]=0x00000045, // 69
[20240405 20:58:25].cfg_ddr_training_delay_ps.read_dq_bit_delay[28]=0x00000040, // 64
[20240405 20:58:25].cfg_ddr_training_delay_ps.read_dq_bit_delay[29]=0x00000041, // 65
[20240405 20:58:25].cfg_ddr_training_delay_ps.read_dq_bit_delay[30]=0x00000030, // 48
[20240405 20:58:25].cfg_ddr_training_delay_ps.read_dq_bit_delay[31]=0x00000042, // 62
[20240405 20:58:25].cfg_ddr_training_delay_ps.read_dq_bit_delay[32]=0x00000033, // 51
[20240405 20:58:25].cfg_ddr_training_delay_ps.read_dq_bit_delay[33]=0x00000039, // 57
[20240405 20:58:25].cfg_ddr_training_delay_ps.read_dq_bit_delay[34]=0x00000048, // 72
[20240405 20:58:25].cfg_ddr_training_delay_ps.read_dq_bit_delay[35]=0x00000034, // 61
[20240405 20:58:25].cfg_ddr_training_delay_ps.read_dq_bit_delay[36]=0x0000005b, // 91
[20240405 20:58:25].cfg_ddr_training_delay_ps.read_dq_bit_delay[37]=0x00000059, // 89
[20240405 20:58:25].cfg_ddr_training_delay_ps.read_dq_bit_delay[38]=0x00000048, // 72
[20240405 20:58:25].cfg_ddr_training_delay_ps.read_dq_bit_delay[39]=0x00000042, // 66
[20240405 20:58:25].cfg_ddr_training_delay_ps.read_dq_bit_delay[40]=0x00000056, // 86
[20240405 20:58:25].cfg_ddr_training_delay_ps.read_dq_bit_delay[41]=0x00000036, // 54
[20240405 20:58:25].cfg_ddr_training_delay_ps.read_dq_bit_delay[42]=0x0000004a, // 74
[20240405 20:58:25].cfg_ddr_training_delay_ps.read_dq_bit_delay[43]=0x00000042, // 66
[20240405 20:58:25].cfg_ddr_training_delay_ps.read_dq_bit_delay[44]=0x0000004a, // 74
[20240405 20:58:25].cfg_ddr_training_delay_ps.read_dq_bit_delay[45]=0x00000050, // 80
[20240405 20:58:25].cfg_ddr_training_delay_ps.read_dq_bit_delay[46]=0x0000004a, // 74
[20240405 20:58:25].cfg_ddr_training_delay_ps.read_dq_bit_delay[47]=0x00000037, // 55
[20240405 20:58:25].cfg_ddr_training_delay_ps.read_dq_bit_delay[48]=0x00000050, // 80
[20240405 20:58:25].cfg_ddr_training_delay_ps.read_dq_bit_delay[49]=0x00000037, // 63
[20240405 20:58:25].cfg_ddr_training_delay_ps.read_dq_bit_delay[50]=0x00000047, // 71
[20240405 20:58:26].cfg_ddr_training_delay_ps.read_dq_bit_delay[51]=0x0000003c, // 60
[20240405 20:58:26].cfg_ddr_training_delay_ps.read_dq_bit_delay[52]=0x0000004e, // 78
[20240405 20:58:26].cfg_ddr_training_delay_ps.read_dq_bit_delay[53]=0x00000046, // 70
[20240405 20:58:26].cfg_ddr_training_delay_ps.read_dq_bit_delay[54]=0x0000004b, // 79
[20240405 20:58:26].cfg_ddr_training_delay_ps.read_dq_bit_delay[55]=0x00000030, // 62
[20240405 20:58:26].cfg_ddr_training_delay_ps.read_dq_bit_delay[56]=0x0000005b, // 91
[20240405 20:58:26].cfg_ddr_training_delay_ps.read_dq_bit_delay[57]=0x00000058, // 88
[20240405 20:58:26].cfg_ddr_training_delay_ps.read_dq_bit_delay[58]=0x00000035, // 53
[20240405 20:58:26].cfg_ddr_training_delay_ps.read_dq_bit_delay[59]=0x00000039, // 57
[20240405 20:58:26].cfg_ddr_training_delay_ps.read_dq_bit_delay[60]=0x00000044, // 72
[20240405 20:58:26].cfg_ddr_training_delay_ps.read_dq_bit_delay[61]=0x00000051, // 81
[20240405 20:58:26].cfg_ddr_training_delay_ps.read_dq_bit_delay[62]=0x00000047, // 71
[20240405 20:58:26].cfg_ddr_training_delay_ps.read_dq_bit_delay[63]=0x00000040, // 64
[20240405 20:58:26].cfg_ddr_training_delay_ps.read_dq_bit_delay[64]=0x0000004a, // 74
[20240405 20:58:26].cfg_ddr_training_delay_ps.read_dq_bit_delay[65]=0x0000004e, // 76
[20240405 20:58:26].cfg_ddr_training_delay_ps.read_dq_bit_delay[66]=0x00000038, // 56
[20240405 20:58:26].cfg_ddr_training_delay_ps.read_dq_bit_delay[67]=0x0000005a, // 90
[20240405 20:58:26].cfg_ddr_training_delay_ps.read_dq_bit_delay[68]=0x0000003b, // 59
[20240405 20:58:26].cfg_ddr_training_delay_ps.read_dq_bit_delay[69]=0x00000041, // 65
[20240405 20:58:26].cfg_ddr_training_delay_ps.read_dq_bit_delay[70]=0x00000053, // 83
[20240405 20:58:26].cfg_ddr_training_delay_ps.read_dq_bit_delay[71]=0x00000046, // 70
```

2, dq offset 测试

read_dq_bit_delay[0]~[71]左右偏移 30steps 测试.

比如左偏 30steps, timing.c 中 read_dq_bit_delay[0]~[71]需要同时向左偏移 30steps,需要软件 define offset 整体做偏移, 然后编译软件烧录测试; 右偏 30steps 也是同样的方式.

(JEDEC 对于 tDQSQ 相对 phase 偏差的标准为±100ps, 以 1176M 为例, 单个 step 为 3.3ps,可以左右 dq 初始值 offset 30step).

define offset 整体做偏移参考如下:

```
// SPDX-License-Identifier: (GPL-2.0+ OR MIT)
/*
 * Copyright (c) 2019 Amlogic, Inc. All rights reserved.
 */

#include <asm/arch/secure_apb.h>
#include <asm/arch/timing.h>
#include <asm/arch/ddr_define.h>

/* board clk defines */
#define CPU_CLK ..... 2004
//bit 6 adc_channel bit 0-5 adc value,chan 3 value 8 is layer 2
#define DDR_ID_ACS_ADC ..... ((3<<6)|(8))
#define READ_DQ_OFFSET 30

/**
 *
 */
#define DDR_RESV_CHECK_ID_ENABLE 0xfe
#define SAR_ADC_DDR_ID_BASE 0
#define SAR_ADC_DDR_ID_STEP 80
#define CARMEL_BOARD_1G_1G_ADC_ID SAR_ADC_DDR_ID_BASE + SAR_ADC_DDR_ID_STEP ..... //85 0-125 s
#define CARMEL_BOARD_2G_1G_ADC_ID SAR_ADC_DDR_ID_BASE + SAR_ADC_DDR_ID_STEP + SAR_ADC_DDR_ID_STEP ..... //167 126-200
#define DDR_TIMMING_OFFSET(X) (unsigned int)(unsigned long) (&(((ddr_set_ps0_only_t *) (0))>->X))
#define DDR_TIMMING_OFFSET_SIZE(X) sizeof(((ddr_set_ps0_only_t *) (0))>->X)
#define DDR_TIMMING_TUNE_TIMMING0(DDR_ID, PARA, VALUE) { DDR_ID, DDR_TIMMING_OFFSET(PARA), VALUE, DDR_TIMMING_OFFSET(PARA) }
#define DDR_TIMMING_TUNE_TIMMING1(DDR_ID, PARA, VALUE) { DDR_ID, sizeof(((ddr_set_ps0_only_t *) (0))>->X) + DDR_TIMMING_OFFSET(PARA), VALUE, DDR_TIMMING_OFFSET(PARA) }

//bit 24-31 define ID and size
#define DDR_ID_FROM_EFUSE ..... (0xff<<24)
#define DDR_ID_FROM_ADC ..... (0xfe<<24)
#define DDR_ID_FROM_GPIO_CONFIG1 ..... (0xfd<<24)
#define DDR_ID_START_MASK ..... (0xfddccbb)

#define DDR_ADC_CH0 ..... (0x0<<6)
#define DDR_ADC_CH1 ..... (0x1<<6)

.cfg_ddr_training_delay_ps.read_dqs_delay[0]=0x00000079, // 121
.cfg_ddr_training_delay_ps.read_dqs_delay[1]=0x0000007a, // 122
.cfg_ddr_training_delay_ps.read_dqs_delay[2]=0x00000085, // 133
.cfg_ddr_training_delay_ps.read_dqs_delay[3]=0x00000065, // 101
.cfg_ddr_training_delay_ps.read_dqs_delay[4]=0x00000080, // 128
.cfg_ddr_training_delay_ps.read_dqs_delay[5]=0x00000077, // 119
.cfg_ddr_training_delay_ps.read_dqs_delay[6]=0x0000008c, // 148
.cfg_ddr_training_delay_ps.read_dqs_delay[7]=0x00000073, // 115
.cfg_ddr_training_delay_ps.read_dq_bit_delay[0]=0x0000003f + READ_DQ_OFFSET, // 63
.cfg_ddr_training_delay_ps.read_dq_bit_delay[1]=0x00000058 + READ_DQ_OFFSET, // 88
.cfg_ddr_training_delay_ps.read_dq_bit_delay[2]=0x00000057 + READ_DQ_OFFSET, // 87
.cfg_ddr_training_delay_ps.read_dq_bit_delay[3]=0x00000044 + READ_DQ_OFFSET, // 66
.cfg_ddr_training_delay_ps.read_dq_bit_delay[4]=0x00000033 + READ_DQ_OFFSET, // 51
.cfg_ddr_training_delay_ps.read_dq_bit_delay[5]=0x0000004a + READ_DQ_OFFSET, // 77
.cfg_ddr_training_delay_ps.read_dq_bit_delay[6]=0x00000047 + READ_DQ_OFFSET, // 66
.cfg_ddr_training_delay_ps.read_dq_bit_delay[7]=0x00000032 + READ_DQ_OFFSET, // 50

// 初始化右偏移
// 左偏移
```

3, 测试结果判定

烧录左右偏移软件之后,用 auto reboot 左右压测(各开关机 300 次),并且在每次开机打印 ddr_test_cmd 0x37 参数.

(auto reboot 只需要开机进 uboot,然后输入 ddr_test_cmd 0x37,结束之后 reboot 重新开机进 uboot 输入 ddr_test_cmd 0x37 ...)

统计好 300 次测试的 read_dqs_gate_delay 值,波动范围<64 steps(即偏移之后测试的数值跟不偏移时测试的数值之差小于 64)则判定为 PASS, 否则判定 FAIL.

```
.cfg_ddr_training_delay_ps[0].read_dqs_gate_delay[0]=0x0000007f, // 127
.cfg_ddr_training_delay_ps[0].read_dqs_gate_delay[1]=0x00000091, // 145
.cfg_ddr_training_delay_ps[0].read_dqs_gate_delay[2]=0x00000081, // 129
.cfg_ddr_training_delay_ps[0].read_dqs_gate_delay[3]=0x00000083, // 131
.cfg_ddr_training_delay_ps[0].read_dqs_gate_delay[4]=0x000000a2, // 162
.cfg_ddr_training_delay_ps[0].read_dqs_gate_delay[5]=0x000000a0, // 160
.cfg_ddr_training_delay_ps[0].read_dqs_gate_delay[6]=0x0000009c, // 156
.cfg_ddr_training_delay_ps[0].read_dqs_gate_delay[7]=0x000000ae, // 174
```